

Linear Regression

Modelo: $p(y|\mathbf{x}, \mathbf{w}) = \mathcal{N}(y|\mathbf{w}^\top \mathbf{x}, \sigma^2)$
Predicción: $\hat{y} = \mathbb{E}[y|\mathbf{x}, \mathbf{w}] = \mathbf{w}^\top \mathbf{x}$
 $\mathcal{L} = \text{MSE} = \frac{1}{N} \sum_i (y_i - \mathbf{w}^\top \mathbf{x}_i)^2$
 $\nabla_{\mathbf{w}} \mathcal{L} = \frac{2}{N} \sum_i (\mathbf{w}^\top \mathbf{x}_i - y_i) \mathbf{x}_i = \frac{2}{N} \mathbf{X}^\top (\mathbf{X} \mathbf{w} - \mathbf{y})$
Solución (MLE / LS): $\mathbf{w}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$

Logistic Regression

Binaria
Modelo: $p(y|\mathbf{x}, \boldsymbol{\theta}) = \text{Ber}(y|\sigma(\mathbf{w}^\top \mathbf{x} + b))$
Predicción (prob): $\mu_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$
 $\mathcal{L} = \frac{1}{N} \sum_i \mathbb{H}_{ce}(y_i, \mu_i)$
 $\nabla_{\mathbf{w}} \mathcal{L} = \frac{1}{N} \sum_i (\mu_i - y_i) \mathbf{x}_i = \frac{1}{N} \mathbf{X}^\top (\boldsymbol{\mu} - \mathbf{y})$
Multiclase
Modelo: $p(y|\mathbf{x}, \mathbf{w}) = \text{Cat}(y|\text{Softmax}(\mathbf{w}^\top \mathbf{x}))$
Predicción (prob): $\boldsymbol{\mu}_i = \text{Softmax}(\mathbf{w}^\top \mathbf{x}_i)$
 $\mathcal{L} = \frac{1}{N} \sum_i \mathbb{H}_{ce}(\mathbf{y}_i, \boldsymbol{\mu}_i)$
 $\nabla_{\mathbf{w}} \mathcal{L} = \frac{1}{N} \sum_i \mathbf{x}_i (\boldsymbol{\mu}_i - \mathbf{y}_i)^\top = \frac{1}{N} \mathbf{X}^\top (\mathbf{M} - \mathbf{Y})$
($\mathbf{M}$ matriz de predicciones, $\mathbf{Y}$ one-hots)

GDA

Modelo: $p(y = c|\mathbf{x}, \mathbf{w}) \propto \pi_c p(\mathbf{x}|y = c, \mathbf{w})$
Class-cond: $p(\mathbf{x}|y = c, \mathbf{w}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$
Prior: $\pi_c = p(y = c|\mathbf{w})$
Entrenamiento: MLE ($\hat{\pi}, \hat{\boldsymbol{\mu}}_c, \hat{\boldsymbol{\Sigma}}_c$)
MAP: $\hat{\boldsymbol{\Sigma}}_{\text{map}} = \lambda \text{diag}(\hat{\boldsymbol{\Sigma}}) + (1 - \lambda) \hat{\boldsymbol{\Sigma}}$
QDA ($\boldsymbol{\Sigma}_c$ libre por clase)
Frontera: $\mathbf{x}^\top \mathbf{A} \mathbf{x} + \mathbf{b}^\top \mathbf{x} + c_0 = 0$
 $\mathbf{A} = \frac{1}{2} (\boldsymbol{\Sigma}_{c'}^{-1} - \boldsymbol{\Sigma}_c^{-1})$, $\mathbf{b} = \boldsymbol{\Sigma}_c^{-1} \boldsymbol{\mu}_c - \boldsymbol{\Sigma}_{c'}^{-1} \boldsymbol{\mu}_{c'}$
 $c_0 = \log \frac{\pi_{c'}}{\pi_c} - \frac{1}{2} \log \frac{|\boldsymbol{\Sigma}_{c'}|}{|\boldsymbol{\Sigma}_c|} - \frac{1}{2} (\boldsymbol{\mu}_c^\top \boldsymbol{\Sigma}_c^{-1} \boldsymbol{\mu}_c - \boldsymbol{\mu}_{c'}^\top \boldsymbol{\Sigma}_{c'}^{-1} \boldsymbol{\mu}_{c'})$
LDA ($\boldsymbol{\Sigma}_c = \boldsymbol{\Sigma} \forall c$)
Frontera: $\mathbf{w}^\top \mathbf{x} + w_0 = 0$
donde $\mathbf{w} = \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu}_c - \boldsymbol{\mu}_{c'})$
 $w_0 = \log \frac{\pi_{c'}}{\pi_c} - \frac{1}{2} (\boldsymbol{\mu}_c + \boldsymbol{\mu}_{c'})^\top \mathbf{w}$

Naive Bayes

Naive Bayes Assumption: $x_d \perp x_{d'} \mid y = c$
Modelo: $p(y = c|\mathbf{x}) \propto \pi_c \prod_{d=1}^D p(x_d|y = c, \boldsymbol{\theta}_{dc})$
 $p(x_d|y = c, \boldsymbol{\theta}_{dc})$ según tipo de x_d :

- Binaria $\text{Ber}(x_d|\boldsymbol{\theta}_{dc})$
- Catórgica $= \text{Cat}(x_d|\boldsymbol{\theta}_{dc})$
- Real $= \mathcal{N}(x_d|\mu_{dc}, \sigma_{dc}^2)$

Entrenamiento: MLE

CART

$\mathcal{L}(\boldsymbol{\theta}) = \sum_i \ell(y_i, f(\mathbf{x}_i; \boldsymbol{\theta})) = \sum_j^J \sum_{\mathbf{x}_i \in R_j} \ell(y_i, w_j)$
Split (nodo i):
$$\arg \min_j \min_t \frac{|\mathcal{D}_i^L|}{|\mathcal{D}_i|} C^L(\mathcal{D}_i^L) + \frac{|\mathcal{D}_i^R|}{|\mathcal{D}_i|} C^R(\mathcal{D}_i^R)$$

 $C(\mathcal{D}_i)$ Regresión $= \text{MSE}$
 $C(\mathcal{D}_i)$ Clasificación:

- $G_i = \sum_c \hat{\pi}_{ic} (1 - \hat{\pi}_{ic})$
- $H_i = \mathbb{H}(\hat{\boldsymbol{\pi}}) = -\sum_c \hat{\pi}_{ic} \log \hat{\pi}_{ic}$

Multi-layer Perceptron (MLP)

$\mathbf{z}^{(\ell)} = f^{(\ell)}(\mathbf{z}^{(\ell-1)}) = \varphi^{(\ell)}(\mathbf{a}^{(\ell)})$: output capa ℓ
 $\mathbf{a}^{(\ell)} = \mathbf{W}^{(\ell)} \mathbf{z}^{(\ell-1)} + \mathbf{b}^{(\ell)}$: preactivación capa ℓ
Back-propagation
Forward-pass
 $\mathbf{z}^{(0)} \leftarrow \mathbf{x}_i$
for $\ell = 1 : L$ **do**
 $\mathbf{a}^{(\ell)} \leftarrow \mathbf{W}^{(\ell)} \mathbf{z}^{(\ell-1)} + \mathbf{b}^{(\ell)}$
 $\mathbf{z}^{(\ell)} \leftarrow \varphi^{(\ell)}(\mathbf{a}^{(\ell)})$
 $\hat{\mathbf{y}}_i \leftarrow \mathbf{z}^{(L)}$, $\mathcal{L}_i \leftarrow \mathcal{L}_i(\mathbf{y}_i, \hat{\mathbf{y}}_i)$
Backward-pass
 $\delta^{(L)} \leftarrow \varphi'^{(L)}(\mathbf{a}^{(L)}) \odot \frac{\partial \mathcal{L}_i}{\partial \mathbf{y}_i}$
 $\frac{\partial \mathcal{L}_i}{\partial \mathbf{W}^{(L)}} \leftarrow \delta^{(L)} \mathbf{z}^{(L-1)\top}$
for $l = L - 1 : 1$ **do**
 $\delta^{(l)} \leftarrow \varphi'^{(l)}(\mathbf{a}^{(l)}) \odot \mathbf{W}^{(l+1)\top} \delta^{(l+1)}$
 $\frac{\partial \mathcal{L}_i}{\partial \mathbf{W}^{(l)}} \leftarrow \delta^{(l)} \mathbf{z}^{(l-1)\top}$
 $\frac{\partial \mathcal{L}_i}{\partial \mathbf{b}^{(l)}} \leftarrow \delta^{(l)}$

Non-Parametric Methods

KNN-classifier:
 $p(y = c|\mathbf{x}, \mathcal{D}) = \frac{1}{K} \sum_{i \in \mathcal{N}_K(\mathbf{x}, \mathcal{D})} \mathbb{I}(y_i = c)$
Parzen Window Density Estimator:
 $p(\mathbf{x}, \mathcal{D}) = \frac{1}{N} \sum_{i=1}^N \mathcal{K}_h(\mathbf{x}, \mathbf{x}_i)$

Perceptron

$\hat{y} = \mathbb{I}(\mathbf{w}^\top \mathbf{x} + b > 0) \rightarrow$ (determinístico)
Update: $\mathbf{w}_{t+1} \leftarrow \mathbf{w}_t - \eta (\hat{y}_i - y_i) \mathbf{x}_i$

Activation Functions (φ)

Sigmoid: $\sigma(z) = \frac{1}{1+e^{-z}}$; $\sigma' = \sigma(1 - \sigma)$
Tanh: $\tanh' = 1 - \tanh^2$
ReLU: $\max(0, z)$; $\text{ReLU}' = \mathbb{I}[z > 0]$
Leaky ReLU: $\max(\alpha z, z)$, $\alpha \ll 1$
Softplus: $\log(1 + e^z)$
Identity: $\varphi(z) = z$
Softmax: $\sigma(\mathbf{z})_i = e^{z_i} / \sum_j e^{z_j}$

NN Regularization + ∇ vanishing sol.

Weight Decay (ℓ_2): MAP estimation
 $p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{0}, \alpha^2 \mathbf{I})$, $p(\mathbf{b}) = \mathcal{N}(\mathbf{b}|\mathbf{0}, \beta^2 \mathbf{I})$
Dropout: $\theta_{lji} = w_{lji} \epsilon_{li}$, $\epsilon \sim \text{Ber}(1 - p)$
Al predecir: escalar activaciones por $(1 - p)$
Early Stopping: frenar cuando $\uparrow \mathcal{L}_{\text{val}}$
Batch Normalization: normalizar activaciones por mini-batch $\mathcal{B}$
 $\hat{z}^{(\ell)} = \frac{z^{(\ell)} - \boldsymbol{\mu}_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}}$, $\tilde{z}^{(\ell)} = \gamma \hat{z}^{(\ell)} + \beta$
Glorot/Xavier initialization
 $w_{ij} \sim \mathcal{N}(0, \sigma^2)$ con $\sigma^2 = 2/(n_{\text{in}} + n_{\text{out}})$
Activaciones sin saturación: ReLU, Leaky ReLU (evitan saturación)
Gradient clipping: $\mathbf{g} = \min(0, \mathbf{g} \cdot \frac{c}{\|\mathbf{g}\|})$
Batch Normalization / Layer Normalization

Optimization

GD: $\mathbf{w}_{t+1} \leftarrow \mathbf{w}_t - \eta_t \nabla_{\mathbf{w}} \mathcal{L}|_{\mathbf{w}_t}$
Newton: $\mathbf{w}_{t+1} \leftarrow \mathbf{w}_t - \eta_t \mathbf{H}_t^{-1} \nabla_{\mathbf{w}} \mathcal{L}|_{\mathbf{w}_t}$
Momentum:
$$\begin{cases} \mathbf{w}_{t+1} \leftarrow -\eta_t \mathbf{m}_t \\ \mathbf{m}_t \leftarrow \beta \mathbf{m}_{t-1} + \nabla \mathcal{L}_{\mathbf{w}}|_{\mathbf{w}_{t-1}} \end{cases}$$

Learning-rate Schedule:

- Lineal: $\eta_t = (1 - \frac{t}{T}) \eta_0 + (\frac{t}{T}) \eta_T$
- Potencia: $\eta_t = \eta_0 (1 + \frac{t}{s})^c$
- Exponencial: $\eta_t = \eta_0 \cdot c^{(t/s)}$

AdaGrad:
$$\begin{cases} w_{t+1,d} \leftarrow w_{t,d} - \eta \frac{1}{\sqrt{s_{k,d} + \epsilon}} \frac{\partial \mathcal{L}}{\partial w_d} \\ s_{t,d} = \sum_{i=1}^{t-1} (\frac{\partial \mathcal{L}_i}{\partial w_d})^2 \end{cases}$$
ADAM:
$$\begin{cases} w_{t+1,d} \leftarrow w_{t,d} - \eta \frac{\hat{s}_{t,d}}{\sqrt{\hat{r}_{k,d} + \epsilon}} \\ \hat{s}_{t,d} = s_{t,d} / (1 - \beta_{1t}) \\ \hat{r}_{t,d} = r_{t,d} / (1 - \beta_{2t}) \\ s_{t,d} = \beta_1 s_{t-1,d} + (1 - \beta_1) \frac{\partial \mathcal{L}}{\partial w_d} \\ r_{t,d} = \beta_2 r_{t-1,d} + (1 - \beta_2) (\frac{\partial \mathcal{L}}{\partial w_d})^2 \end{cases}$$

Regression Metrics

$e_i \triangleq y_i - \hat{y}_i$; $\text{RSS} \triangleq \sum_i e_i^2$; $\text{TSS} \triangleq \sum_i (y_i - \bar{y})^2$
 $\text{MSE} = \frac{1}{N} \text{RSS}$
 $\text{RMSE} = \sqrt{\text{MSE}}$
 $\text{MAE} = \frac{1}{N} \sum_i |e_i|$
 $R^2 \triangleq 1 - \frac{\text{RSS}}{\text{TSS}} \in [0, 1]$ (varianza explicada)

Classification Metrics

$\text{Acc} = \frac{TP+TN}{TP+TN+FP+FN}$ $\text{Recall} = \frac{TP}{TP+FN}$
 $\text{Precision} = \frac{TP}{TP+FP}$ $\text{FPR} = \frac{FP}{FP+TN}$
 $\text{FDR} = \frac{FP}{FP+TP}$
 $F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP+FP+FN}$
ROC: FPR vs TPR=Recall
PR: Recall vs Precision

Maximum A Posteriori (MAP)

$\mathbf{w}^* = \arg \max_{\mathbf{w}} p(\mathbf{w}|\mathcal{D}) = \arg \max_{\mathbf{w}} p(\mathcal{D}|\mathbf{w}) p(\mathbf{w})$
 $\ell_2 \rightarrow p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{0}, \lambda^{-1} \mathbf{I}) \rightarrow \mathcal{L} + \lambda \|\mathbf{w}\|^2$
 $\ell_1 \rightarrow p(\mathbf{w}) = \text{Laplace}(\mathbf{w}|\mathbf{0}, \lambda^{-1}) \rightarrow \mathcal{L} + \lambda \|\mathbf{w}\|_1$

Bayes

$p(\mathbf{w}|\mathcal{D}) = \frac{p(\mathcal{D}|\mathbf{w}) p(\mathbf{w})}{p(\mathcal{D})} \propto p(\mathcal{D}|\mathbf{w}) p(\mathbf{w})$
posterior $\propto$ *likelihood* $\times$ *prior*

Distributions + MLE estimators

Gaussian $\mathcal{N}(x|\mu, \sigma^2)$:
$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$
$$\hat{\mu} = \frac{1}{N} \sum_i x_i \quad \hat{\sigma}^2 = \frac{1}{N} \sum_i (x_i - \hat{\mu})^2$$
Gaussian Multivariada $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$:
$$\frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})\right)$$
$$\hat{\boldsymbol{\mu}} = \frac{1}{N} \sum_i \mathbf{x}_i \quad \hat{\boldsymbol{\Sigma}} = \frac{1}{N} \sum_i (\mathbf{x}_i - \hat{\boldsymbol{\mu}})(\mathbf{x}_i - \hat{\boldsymbol{\mu}})^\top$$
Bernoulli $\text{Ber}(x|\mu)$:
$$\mu^x (1 - \mu)^{1-x}$$
$$\hat{\mu} = \frac{1}{N} \sum_i x_i$$
Catórgica $\text{Cat}(y|\boldsymbol{\pi})$:
$$\prod_c \pi_c^{\mathbb{I}[y=c]}, \quad \sum_c \pi_c = 1$$
$$\hat{\pi}_c = \frac{N_c}{N} \quad (\text{frecuencia de clase } c)$$
Laplace $\text{Lap}(x|\mu, b)$:
$$\frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right)$$
$$\hat{\mu} = \text{median}(\{x_i\}) \quad \hat{b} = \frac{1}{N} \sum_i |x_i - \hat{\mu}|$$
Binomial $\text{Binom}(k|n, p)$:
$$\binom{n}{k} p^k (1 - p)^{n-k}$$
$$\hat{p} = \frac{1}{N} \sum_i \frac{k_i}{n}$$

Information Theory

Entropy: $\mathbb{H}(\mathbf{p}) = -\sum_i p_i \log p_i$
Cross-E: $\mathbb{H}_{ce}(\mathbf{p}, \mathbf{q}) = -\sum_i p_i \log q_i$
Binary Cross-E:
 $\mathbb{H}_{ce}(p, q) = -[p \log q + (1 - p) \log(1 - q)]$
Joint-E: $\mathbb{H}(X, Y) = -\sum p(x, y) \log p(x, y)$
Conditional-E: $\mathbb{H}(Y|X) = \mathbb{H}(X, Y) - \mathbb{H}(X)$
Relative-E (KL divergence)
 $\mathbb{H}(p||q) = \sum_i p_i \log \frac{p_i}{q_i} = \mathbb{H}(p||q) - \mathbb{H}(p)$
Mutual Information
 $I(X; Y) = \mathbb{H}(p(x, y) || p(x)p(y))$
 $I(X; Y) = 0 \Leftrightarrow X \perp Y$

Linear Algebra

SDP: $\mathbf{A} \succeq 0 \Leftrightarrow \mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0 \forall \mathbf{x}$.
 $\mathbf{A}^\top \mathbf{A}$, $\mathbf{A} \mathbf{A}^\top \succeq 0$ siempre
Cond: $\kappa(\mathbf{A}) = \sigma_{\text{max}}/\sigma_{\text{min}}$, $\kappa(\mathbf{A}^\top \mathbf{A}) = \kappa(\mathbf{A})^2$
Precisión: $\boldsymbol{\Lambda} \triangleq \boldsymbol{\Sigma}^{-1}$
Derivadas conocidas

- $\nabla_{\mathbf{x}}(\mathbf{x}^\top \mathbf{A} \mathbf{x}) = 2\mathbf{A} \mathbf{x}$ ($\mathbf{A}$ simétrica)
- $\nabla_{\mathbf{x}}(\mathbf{a}^\top \mathbf{x}) = \mathbf{a}$
- $\nabla_{\boldsymbol{\Sigma}} \log |\boldsymbol{\Sigma}| = \boldsymbol{\Sigma}^{-\top}$

Traza: $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \text{tr}(\mathbf{A} \mathbf{x} \mathbf{x}^\top)$, $\text{tr}(\mathbf{A} \mathbf{B}) = \text{tr}(\mathbf{B} \mathbf{A})$
Jacobiano: $\mathbf{J} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \in \mathbb{R}^{m \times n}$, $J_{ij} = \frac{\partial f_i}{\partial x_j}$
Si $m = 1 \implies \mathbf{J} = \nabla f^\top$.

Convexity

$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y})$
1er orden: $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})$
2do orden: $\nabla^2 f(\mathbf{x}) \succeq 0 \forall \mathbf{x}$ (Hessiano SDP)

Ensemble Methods

Bagging: bootstrapping aggregating.
Boosting: entrenamiento secuencial, ponderar errores. f_m se entrena sobre residuos de f_{m-1}
Random Forest: Bagging + features bagging por split.